

IAN SEQUEIRA

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Research Interests: ML reliability, actuarial risk modeling, algorithmic fairness, anytime-valid inference

EDUCATION

University of California, Santa Barbara

B.S. Statistics & Data Science, B.A. Economics

Santa Barbara, CA

Sept 2023 – June 2027

- GPA: 3.84 / 4.0 | Dean's Honors: 7 quarters | Graduate coursework: Statistical Machine Learning (A)
- **Selected coursework:** Design of Experiments (A+), Regression Analysis (A), Stochastic Processes (A), Data Science (A), Econometrics (A), Proof Writing (A)

RESEARCH EXPERIENCE

Qu Lab, UCSB Statistics & Applied Probability

Undergraduate Researcher — Prof. Annie Qu (Founding Director, CSFAI)

Santa Barbara, CA

Apr 2026 – Present

- Developing an anytime-valid runtime monitor for multi-agent LLM pipelines that attaches to any per-turn LLM-judge score and provides formal type-I error control at arbitrary stopping times. Targeting ICLR 2027 (submission Sept 25, 2026).
- Methodological work on e-value constructions for rubric-anchored discrete score outputs and role-conditional nulls for multi-agent settings; empirical evaluation shows nominal type-I control with meaningful detection power on held-out benchmark trajectories.

Summer Institute for Biostatistics & Data Science (iSI-BUDS)

Research Fellow — Prof. Dan Gillen & Prof. Josh Grill

UC Irvine, CA

June 2025 – Aug 2025

- Constructed GLM, ridge, and CART classifiers in R to identify Alzheimer's biomarkers (pTau217, PACC, ADL) across 1,150+ A4 preclinical trial participants; best model ROC-AUC 0.835.
- Developed biomarker-guided sample-size framework achieving 80% statistical power with 188 participants vs. standard 500+.
- Contributing co-author on the resulting biomarker-guided trial design study; results appeared as a co-authored poster at CTAD 2025, San Diego (see Publications).

PUBLICATIONS AND PRESENTATIONS

Sequeira, I., Qu, A. Anytime-valid monitoring for multi-agent LLM systems. Manuscript in preparation, targeting ICLR 2027 submission (Sept 25, 2026). 2026

Gaona-Partida, P., Anuurad, T., Sequeira, I., Ragaine, E., Turner, C., Grill, J. D., Gillen, D. L. *Instructing Efficient Early-phase Preclinical Alzheimer's Disease Trials*. Manuscript in preparation. Preliminary results presented as a poster at *Clinical Trials on Alzheimer's Disease (CTAD)*, San Diego, CA, Dec 2025. 2025 – 2026

HONORS AND AWARDS

SIOP Summer Research Scholarship, UCSB Statistics & Applied Probability

(mentor: Prof. Annie Qu)

2026

Hispanic Scholarship Fund (HSF) Scholar (renewable)

2024 – Present

Dean's Honors, UC Santa Barbara (7 quarters)

2023 – 2026

PROFESSIONAL EXPERIENCE

Alida Inc.

Data Engineering Intern

Remote

Sept 2024 – July 2026

- Architected a multi-mode AI data exploration platform on Databricks operating across 47 enterprise datasets ranging from CMS healthcare to finance and shipping, combining multi-agent routing, graph analysis, and

automated SQL generation.

- Rewrote the platform's core join logic to correctly handle nested-array semantics, resolving accuracy failures on CMS-style hierarchical datasets and cutting query complexity by 30%.
- Extended the platform with a spec-driven ML prediction pipeline featuring typed contracts, dual-layer verification, and fault-tolerant LLM orchestration with AutoML training.

Turing

Applied AI/GenAI Software Engineering Intern

Remote

Feb 2026 – May 2026

- Shipped a call-triage system for a real-estate call-center scenario modeled after CBRE operations, using split-conformal prediction to gate LLM routing across a 32-vendor catalog with distribution-free coverage guarantees, plus a rule-based safety validator for emergency escalation.
- Held-out evaluation on 800 calls: zero missed emergencies and full auto-clearance of routine dispatches meeting the confidence threshold, replacing human triage with a statistically calibrated deployment gate rather than heuristic thresholding.

TEACHING EXPERIENCE

Workshop Instructor, ACM at UCSB

2025 – 2026

- Designed and taught a 4-workshop series for the UCSB ACM chapter spanning applied ML from regression through transformer models, plus multi-agent LLM systems and ML security.

SELECTED PROJECTS

Healthcare Financial Toxicity Prediction

Graduate Statistical Machine Learning final project

Mar 2026

- Predicted catastrophic out-of-pocket burden in 20,962 MEPS 2022 respondents; tuned 6 classifiers under 10-fold cross-validation, XGBoost reaching ROC-AUC 0.874.
- Evaluated subgroup fairness across race/ethnicity using DALEX and *fairmodels*; interpreted feature contributions as SHAP odds ratios to surface actionable clinical predictors.

Research Advisor Finder

Full-stack faculty discovery platform (FastAPI, PostgreSQL/pgvector, Next.js)

Nov 2025 – Feb 2026

- Matched student research interests to 15,000+ professors across 12 STEM fields via hybrid semantic search over CV-extracted interest embeddings.

LEADERSHIP AND SERVICE

Project Executive, ACM Industry, UCSB

Sept 2025 – Present

- Leading an 8-person peer-mentorship team delivering applied ML solutions on 12-week industry sprints; program partners include Turing, Raytheon, and Unwrapped.

Finance & Funding Chair, CampMed, UCSB

Sept 2025 – Present

- Leading budget planning and sponsorship strategy for health-equity outreach to underrepresented high school students.

TECHNICAL SKILLS

Programming: R, Python, SQL, C++, TypeScript, LaTeX | **Methods:** e-processes, anytime-valid inference, conformal prediction, GLMs, XGBoost, CART, SHAP, DALEX, Monte Carlo | **Tools:** tidymodels, rstan, scikit-learn, PyTorch, Databricks, LangGraph

PROFESSIONAL MEMBERSHIPS

American Statistical Association (ASA) | Institute of Mathematical Statistics (IMS) | UCSB Innovators Circle